

# Video-of-Thought: Step-by-Step Video Reasoning from Perception to Cognition

Hao Fei, Shengqiong Wu, Wei Ji, Hanwang Zhang, Meishan Zhang, Mong Li Lee, Wynne Hsu

## 01 Limits Of Current Video Understanding

- Current systems handle shallow tasks but fail on complex, long-horizon reasoning
- Lack fine-grained spatial-temporal grounding for precise tracking and interactions
- Weak linkage from pixel observations to cognitive interpretations and causality
- Gaps hinder reliability in real-world video reasoning applications

## 01 Human-Inspired Multi-Step Reasoning



- Humans progress from perception to cognition through staged steps
- Identify targets, ground and track, analyze interactions, apply commonsense
- Structured steps enable precise grounding and explainable conclusions

## 02 Related Work And Gaps In The Literature

### Video MLLMs And CoT Prompting

Video MLLMs improve general understanding but lack fine-grained grounding

Underutilize LLM commonsense for higher-level cognition

Direct CoT with a single prompt yields modest video gains

Need decomposition aligned with spatiotemporal video structure

### Scene Graphs For Spatiotemporal Modeling

Scene graphs capture objects and relations across frames

STSGs aid compositional video semantics and grounding

Integration into MLLMs for end-to-end reasoning is limited

Need models that understand and generate STSGs as rationales

## 03 Contributions At A Glance

- Introduce MotionEpic integrating and generating STSGs for grounding
- Propose Video-of-Thought (VoT) multi-step reasoning pipeline
- Design grounding-aware objectives aligning video and STSG
- Achieve state-of-the-art on eight complex video QA benchmarks

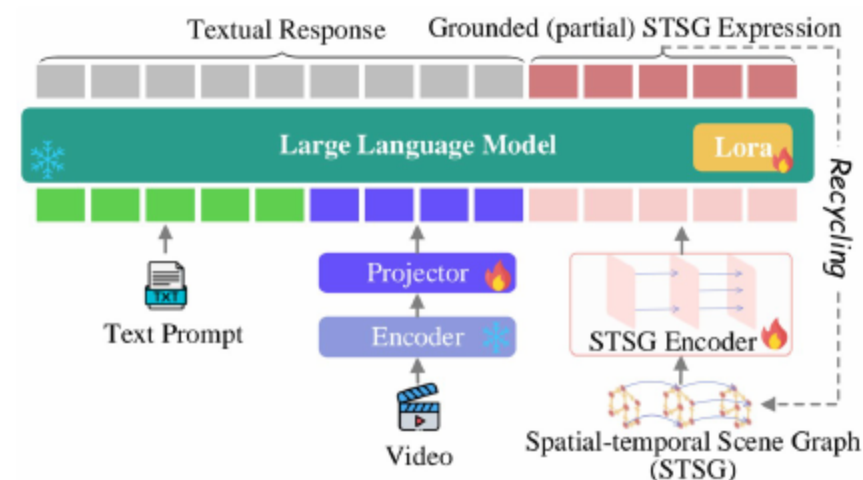
## 03 Narrative Of The Approach

- Architecture and STSG integration enable structured perception
- Grounding-aware tuning teaches autonomous STSG parsing/generation
- VoT decomposes tasks into five verifiable reasoning steps

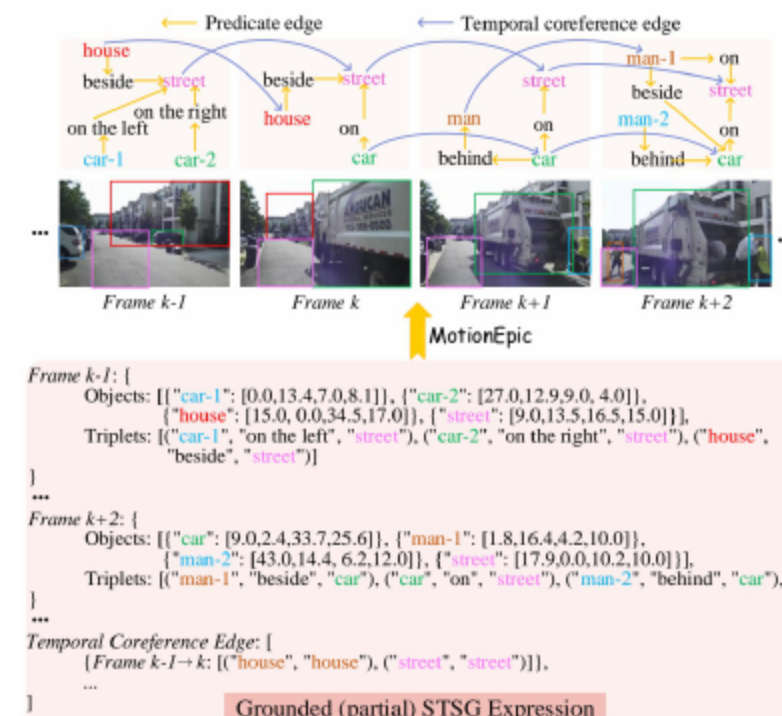


## 04 Architecture Briefing

- Inputs: text, video, and STSG for closed-loop grounding
- Backbones: ViT-L/14 video encoder, Q-Former, Vicuna-7B
- Recurrent Graph Transformer encodes multi-frame STSGs
- Generates STSGs as intermediate rationales for reasoning



## 04 STSG Representation And Extensions

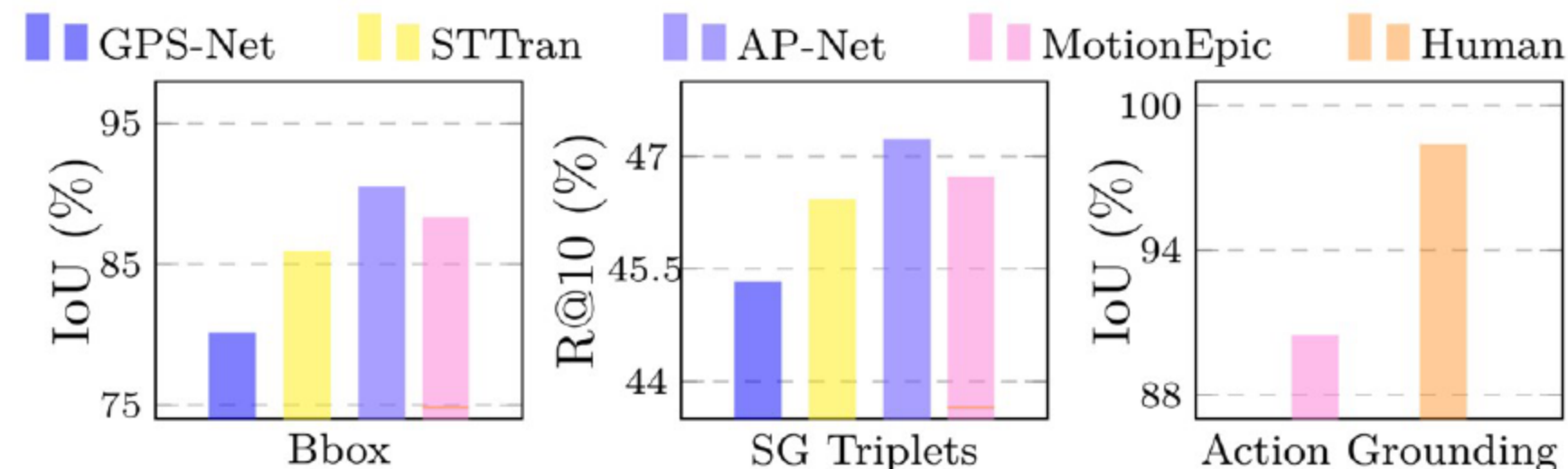


- STSG nodes: objects with category, embedding, and bbox
- Edges: relations plus temporal coreference linking across frames
- Uniform frame sampling balances fidelity and compute
- Generated STSG expressions align and can be recycled

## 05 Coarse-Grained Correspondence Objectives

- L1: predict video-STSG pair matching for global alignment
- L2: generate full STSG expression directly from video
- Enables parsing structured semantics without extra annotations

## 05 Fine-Grained Correspondence Objectives

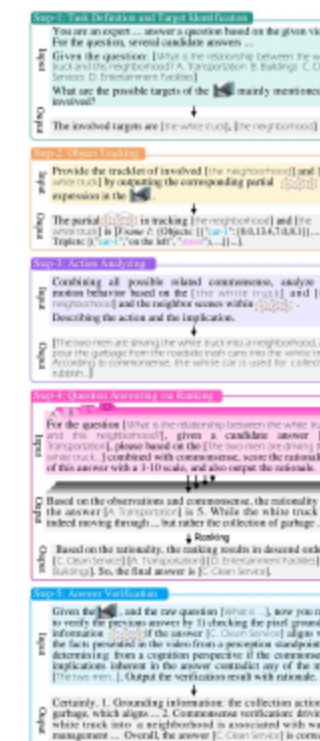


- L3: output target tracklets from video and action text
- L4: describe actions and produce tracklets given key objects
- L5: predict object label and tracklet from single-frame bbox

## 05 Training Protocol And Implementation

- Pretrain on WebVid; then grounding-aware tuning with Action Genome
- Video instruction tuning using VideoChat and Video-ChatGPT
- Freeze encoders mostly; adapt LLM via LoRA
- Sample at 8 fps for coverage and compute balance

## 06 Step 1: Task Definition And Target Identification

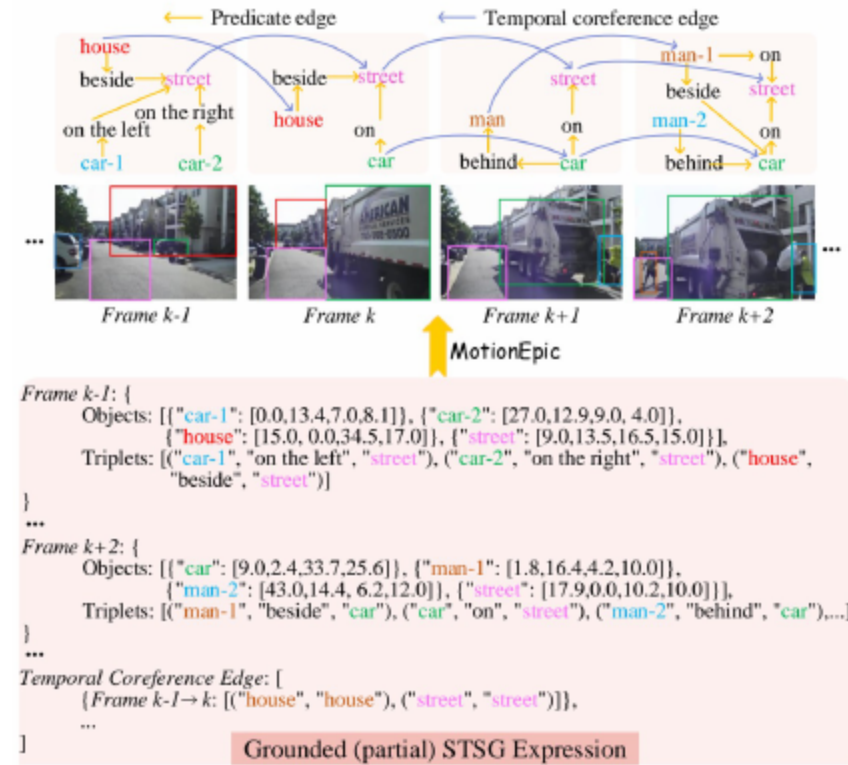


- Identify likely targets relevant to the query
- Prompting focuses perception on pertinent entities
- Establishes basis for subsequent grounding and analysis



## 06 Step 2: Object Tracking Via STSG

- Produce STSG tracklets as precise spatiotemporal evidence
- Mitigates hallucinations and yields reliable trajectories
- Serves as low-level rationale for action interpretation



## 06 Step 3: Action Analysis With Commonsense

Question (a): How many people are wearing white clothes?

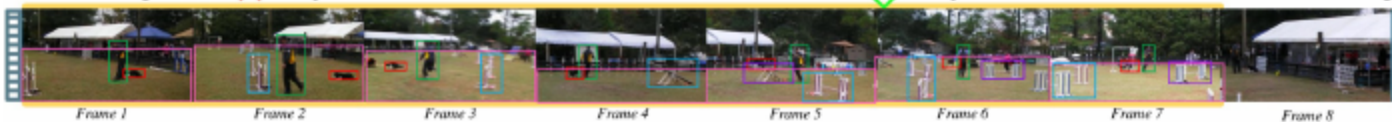


- Integrate tracklets and context, augment with commonsense
- Infer behaviors, interactions, and implications beyond pixels
- Output Observation and Implication summary for cognition

## 06 Step 4: Answering By Candidate Ranking

Question: What is the least likely category for the animal in this video?

A. Police Dog B. Competitive Animal C. Circus Performer D. Companion Pet E. Search and Rescue Dog



- Score candidates 1-10 using observations and commonsense
- Rank and select the most plausible answer
- Provide concise justification aligned with inferred semantics

## 06 Step 5: Dual Verification For Reliability



- Verify grounding consistency with evidence and tracklets
- Check cognitive consistency with commonsense observations
- Conflicts trigger re-ranking for coherent decisions

## 07 Benchmarks And Settings

| Model            | VLEP        | STAR        |             |             |             | IntentQA    | Social-IQ   |             |
|------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
|                  |             | Int.        | Seq.        | Pre.        | Fea.        |             | 2-Way       | 4-Way       |
| ● SoTA baselines |             |             |             |             |             |             |             |             |
| InternVideo      | 63.9        | 62.7        | 65.6        | 54.9        | 51.9        | -           | -           | -           |
| LLaMA-VQA        | <u>71.0</u> | 66.2        | 67.9        | 57.2        | 52.7        | -           | -           | -           |
| VLAP             | 69.6        | <u>70.0</u> | <u>70.4</u> | <u>65.9</u> | <u>62.2</u> | -           | -           | -           |
| SeViLA           | 68.9        | 63.7        | 70.4        | 63.1        | 62.4        | -           | -           | -           |
| VideoChat        | 62.0        | 63.2        | 66.8        | 54.1        | 49.6        | 59.3        | 67.7        | 37.8        |
| Video-LLaVA      | 65.8        | 64.3        | 67.0        | 56.5        | 50.1        | 62.5        | 68.9        | 39.2        |
| ● CoT            |             |             |             |             |             |             |             |             |
| Video-LLaVA      | 65.7        | 65.0        | 67.7        | 57.8        | 52.0        | 63.2        | 69.5        | 40.4        |
| Video-LLaVA+STSG | 67.0        | 65.9        | 68.9        | 58.7        | 53.7        | 64.9        | 70.4        | 41.7        |
| MotionEpic       | 68.2        | 66.8        | 69.6        | 60.6        | 57.4        | <u>66.1</u> | <u>71.7</u> | <u>43.0</u> |
| ● VoT            |             |             |             |             |             |             |             |             |
| MotionEpic       | <b>73.4</b> | <b>71.5</b> | <b>72.6</b> | <b>66.6</b> | <b>62.7</b> | <b>70.8</b> | <b>72.8</b> | <b>45.0</b> |

- Evaluate on VLEP, STAR, IntentQA, Social-IQ, CausalVidQA, NExT-QA
- Zero-shot on MSR-VTT, ActivityNet, NExT-QA, STAR
- Accuracy compared against strong MLLMs and SoTAs

## 08 Finetuned Performance And Comparisons

| Model            | Acc@D       | Acc@E       | Acc@P       |             |             | Acc@C       |             |             |
|------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
|                  |             |             | A           | R           | AR          | A           | R           | AR          |
| ● SoTA baselines |             |             |             |             |             |             |             |             |
| TranSTR          | 73.6        | 75.8        | 65.1        | 65.0        | 48.9        | 68.6        | 65.3        | 50.3        |
| Video-LLaMA      | 69.2        | 71.0        | 63.6        | 62.4        | 44.4        | 65.4        | 60.1        | 45.0        |
| VideoChat        | 72.9        | 73.9        | 65.2        | 63.1        | 45.9        | 66.0        | 62.7        | 45.8        |
| Video-ChatGPT    | 73.1        | 75.1        | 66.0        | 63.9        | 46.0        | 67.8        | 63.6        | 50.0        |
| Video-LLaVA      | 73.7        | 74.4        | 67.6        | 65.4        | 47.7        | 68.0        | 64.9        | 51.5        |
| ● CoT            |             |             |             |             |             |             |             |             |
| Video-LLaVA      | 74.2        | 74.8        | 68.0        | 65.7        | 48.1        | 70.3        | 65.7        | 52.9        |
| Video-LLaVA+STSG | 75.7        | 75.9        | 68.9        | 67.2        | 50.0        | 70.7        | 67.2        | 53.6        |
| MotionEpic       | <u>78.5</u> | <u>77.2</u> | <u>70.1</u> | <u>70.8</u> | <u>52.4</u> | <u>71.2</u> | <u>69.1</u> | <u>55.0</u> |
| ● VoT            |             |             |             |             |             |             |             |             |
| MotionEpic       | <b>81.2</b> | <b>83.0</b> | <b>74.3</b> | <b>73.7</b> | <b>54.7</b> | <b>74.5</b> | <b>73.8</b> | <b>58.6</b> |

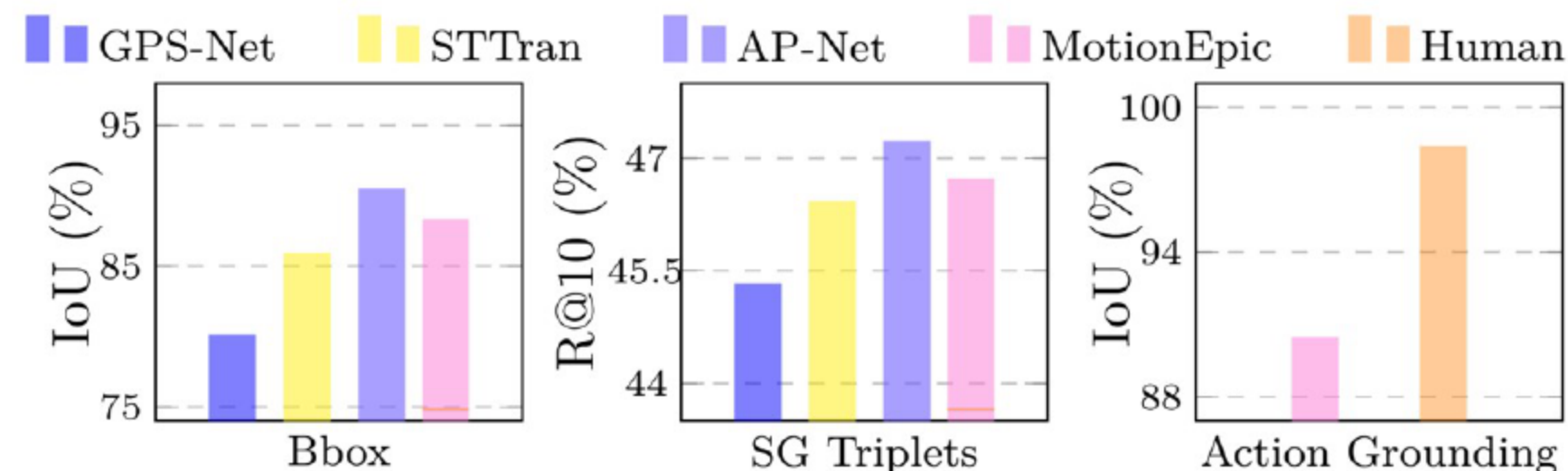
- MotionEpic+VoT sets new state-of-the-art across benchmarks
- STSG integration consistently improves performance
- VoT outperforms vanilla CoT by large margins

## 08 Zero-Shot Improvements And Verification Effects

| Model                      | MSR-VTT     | ActivityNet | NExT-QA     | STAR        | AVG.        |
|----------------------------|-------------|-------------|-------------|-------------|-------------|
| • Zero-shot SoTA baselines |             |             |             |             |             |
| InternVideo                | -           | -           | 49.1        | 41.6        | -           |
| Video-LLaMA                | 49.6        | 21.4        | 43.5        | 36.4        | 37.7        |
| VideoChat                  | 52.0        | 26.5        | 52.8        | 45.0        | 44.1        |
| Video-ChatGPT              | 54.3        | 35.2        | 53.0        | 48.7        | 47.8        |
| Video-LLaVA                | 59.2        | 45.3        | 57.3        | 50.6        | 53.1        |
| VideoChat2                 | 54.1        | 49.1        | 61.7        | <u>59.0</u> | 56.0        |
| • CoT                      |             |             |             |             |             |
| Video-LLaVA                | 60.0        | 46.9        | 59.5        | 52.0        | 54.6        |
| Video-LLaVA+STSG           | 61.5        | 48.4        | 60.6        | 52.7        | 55.8        |
| MotionEpic                 | <u>63.1</u> | <u>50.0</u> | <u>61.9</u> | 56.5        | <u>57.8</u> |
| • VoT                      |             |             |             |             |             |
| MotionEpic                 | <b>66.2</b> | <b>54.6</b> | <b>66.5</b> | <b>61.7</b> | <b>62.3</b> |
| w/o Verify-G               | 63.6        | 51.4        | 62.0        | 59.1        | 59.0        |
| w/o Verify-C               | 65.1        | 53.4        | 62.8        | 58.8        | 60.1        |

- VoT yields larger zero-shot gains than CoT
- Ablating grounding or verification reduces performance
- Perceptual checks crucial for descriptive tasks; both for complex reasoning

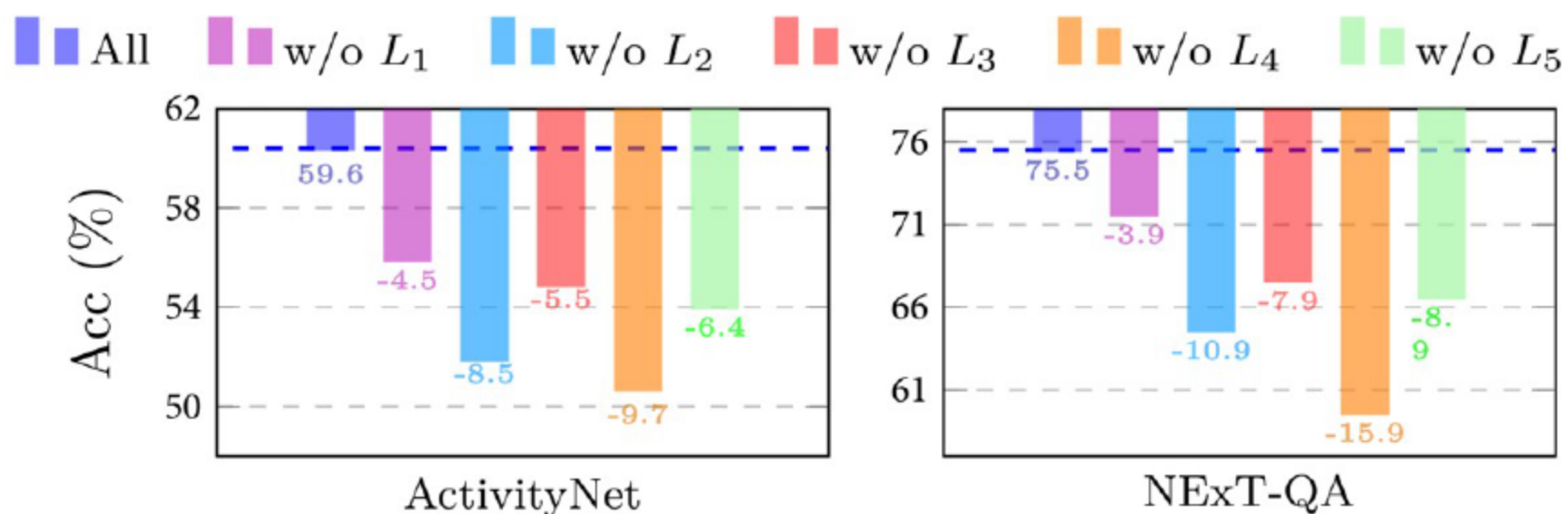
## 09 Probing Video Grounding Ability



- Competitive object grounding and triplet classification
- Strong temporal action grounding vs. DSG parsers
- Validates reliable grounding rationales for reasoning

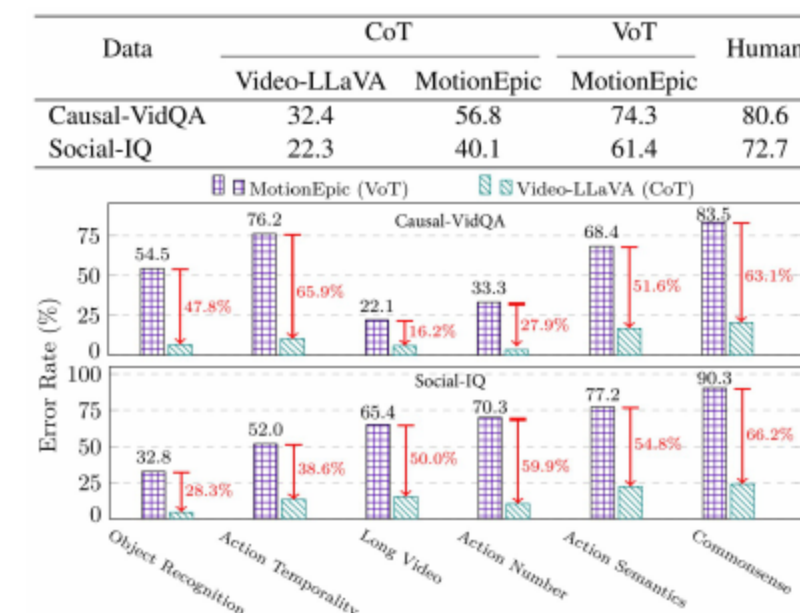


## 09 Impact Of Grounding Objectives



- Ablations show varied drops across objectives
- $L_2$  and  $L_4$  are most critical for performance
- Tight coupling of perception and reasoning is necessary

## 09 Error Analysis And Human Comparison



- Approaches human performance on hardest subsets
- Reduces errors in action semantics and commonsense
- Qualitative studies show grounded, explainable decisions

## 10 Summary Of Findings

- Coupling STSG grounding with VoT advances perception-to-cognition
- Achieves SoTA on complex video QA in finetuned and zero-shot
- Provides reliable, explainable grounding rationales



## 10 Outlook And Extensions

- Extend VoT to forecasting, counterfactuals, safety, interactive agents
- Pursue richer temporal abstractions and broader commonsense
- Improve efficiency for long videos and scalable verification

The image features a minimalist design on a white background. A central white parallelogram with a teal border contains the text 'THANKS!'. This parallelogram is flanked by two teal rectangles. In the top-left corner, there are two overlapping teal triangles. In the top-center, there are two thin teal triangles pointing downwards. In the bottom-right, there is a teal triangle pointing left towards the central shape. A thin black horizontal line runs across the middle of the image, and a thin black vertical line runs down the right side, intersecting at the center.

**THANKS!**